



M.UMAIR RAZA

Aerosystems | Avionics Systems Engineer
Systems Integration Lead

CSEP | PMP | PMI-ACP

ABOUT ME

Systems Engineering professional with 15+ years of experience across aerospace and defense programs, spanning avionics & mission system design, integration, qualification, certification, airworthiness, production, and lifecycle sustainment. Extensive background in MBSE-driven techniques and systems integration. International Proven experience delivering Aerospace and technology retrofit and upgrade programs, enabling multi-country exports, and achieving high cost and lifecycle savings through indigenization and in-country technology development. **PMP and PMI-ACP** certified to bring a systems-thinking mindset to complex, mission-critical environments.

EDUCATION

Bachelor of Avionics Engineering

2004 - 2008

- College of Aeronautical Engineering, Risalpur

Master's in Avionics Engineering

2013 - 2015

- Air University, Islamabad

LICENSES &

CERTIFICATIONS

Project Management Institute, USA

- Project Management Professional
- Agile Certified Practitioner (ACP)

Pakistan Engineering Council

- Professional Engineer (PE)

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📍 Jeddah, Saudi Arabia

🌐 www.linkedin.com/in/mumairaza

EXPERIENCE (15 YRS)

Program Lead, Professional Education (2023 - Present)

College of Aeronautical Engineering & NUTECH University

- Leadership in transformative programs (Capacity Building initiative by the Higher Education Commission, Pakistan)
- Led the integration of 60+ Aerospace and defense projects across multi-disciplinary engineering teams.
- Strong product management in Data, AI, AI/ML lifecycle, analytics, Power BI (KPIs, ROIs).

Dy Director, Program Management Office (2020-23)

Air Force PMO for Systems Integration, Flight Tests and Operations

- Led configuration management and integration of Aero Systems across fixed-wing and UAV platforms; conducted FMECA, LORA, and RCM analyses that improved fleet serviceability to >90%.
- Delivered hands-on field leadership across airworthiness compliance, lifecycle supportability, system integration, and testing through flight-test phases, and operational readiness
- Wrote Risk mitigation plans, RAID logs, and EVM reporting

Lead Systems Integration Expert (2018-20)

Chengdu Aircraft Design Institute, China

- Led requirements & stakeholder analysis to translate Air Force operational needs. Developed and managed Acceptance Test Requirements (ATRs) and verification/test specifications.
- Executed integration and validation testing of avionics subsystems on an advanced aircraft design/integration rig.

Maintenance Support Engineer (AEW&C) (2016-17)

Airborne Warning & Control Systems Engineer for the Air Force

- Delivered flight-line maintenance and sustainment support - fault isolation, real-time troubleshooting, and corrective action planning.
- Performed reliability (MMTR, MTTF), and trend analysis, optimize staggered and preventive maintenance plans, and improved mission availability and aircraft uptime.

Integrated Systems & Design Specialist (2008 -13)

Pakistan Aeronautical Complex Kamra, Pakistan

- Designed, developed, and qualified a mission-critical aircraft subsystems; successfully integrated into operational fleet.
- Led the design, planning, and upgrade of the country's first legacy trainer aircraft with a glass cockpit, enhancing pilot training capabilities to align with advanced fighter platforms.

CORE COMPETENCIES

- **Systems Integration.** Integrated complex avionics and weapons subsystems across aircraft platforms using MBSE (Cameo), with system modeling, interface definition, & model based verification & simulation (MATLAB).
- **Verification & Validation (V&V).** Led end-to-end V&V activities including ground and flight testing, acceptance, and airworthiness compliance.
- **Requirements & Configuration Management.** Executed end-to-end requirements Engineering using IBM Rational DOORS. Enforced configuration baselines, change control, and impact analysis to maintain system integrity.
- **Lifecycle & Fleet Sustainment Engineering.** Drove fleet readiness and sustainment strategies through data-driven reliability analysis, maintenance planning, and configuration governance, improving availability and long-term system supportability.
- **Program Execution & Quality Assurance.** Delivered complex programs, structured V&V gates, and issue/risk tracking (Jira) to ensure compliance with MIL-STD, DO-178C, and DO-254 standards.
- **Cross-Functional Leadership & Global Collaboration.** Led multidisciplinary, cross-cultural engineering teams in joint-development programs across China and Pakistan.

PROJECTS OF NATIONAL IMPORTANCE

Sub-System Development

Leading Fighter Program

Indigenous Development of Backup Mission Computer

- Over **135 systems produced** and deployed on in-service aircraft, eliminating reliance on imports and generating **1.27 million-dollar** cost savings for the Air Force.

Development of a Novel ECCM Algorithm for Radars

- Developed a novel ECCM algorithm for ground-based legacy radars, enhancing clutter rejection and resistance to noise jamming; successfully deployed on in-service MPDR Radar systems.

Glass Cockpit Modification

Trainer Aircraft

Upgrade of Trainer Aircraft with Dynon and Garmin Suite:

- Glass cockpit Modified aircraft exported to **11 countries**, including **Saudi Arabia, Turkey, Qatar**, among others, **11.6 million dollars** in revenue for the Air Force.

Development of Configuration Management System

- Led configuration management for advanced aircraft undergoing simultaneous development and deployment, ensuring controlled evolution and mission support

ACHIEVEMENTS

AWARDS & RECOGNITIONS

- **Chief of the Airstaff Commendation:** for exceptional Services during the avionics integration and retrofit Program
 - Appreciation letter from the Chairman of Pakistan Aeronautical Complex
- **Chengdu Aircraft Design Institute (CADI), China:** Commendation for Development & Integration activities of Lead Fighter Aircraft Program
- **Higher Education Commission (HEC), Pakistan:** Selected for the National Capacity Building Program in System Engineering and allied disciplines

TRAINING COURSES & WORKSHOPS

- **Transfer of Source Code (Aircraft) Training (2018-20):** Chengdu Aircraft Design Institute, China
- **System Engineering Specialization:** University of Colorado Boulder, USA
- **Engineering Management Specialization:** Rice University, USA
- **Project Management Specialization:** Google
- Fighter Aircraft & AWACS Aircraft Type Rating Courses
- Basic Staff Course
- Senior Command & Staff Course

PROFILE(S)

- Coursera: <https://www.coursera.org/learner/umair-raza-4894>